

Fixational Eye Movements in Visual, Cognitive, and Movement Disorders



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Abstract The eyes never remain completely still. Even when attempting to maintain our gaze stable looking at a small target, fixational eye movements keep the eyes in constant motion. Given the simplicity of the task and the rich dataset that can be obtained, the interest in fixational eye movements has grown in multiple research and clinical fields. First, this chapter reviews the general classes of eye movements and how they may contribute to the main two components of fixational eye movements: drifts and microsaccades. While microsaccades are considered to be part of a continuum with saccades, drift results from a combination of all other smooth eye movements, such as vestibular ocular reflex, smooth pursuit, and vergence. Then, it discusses the methods used to analyze fixational eye movements and what are the typical parameters of interest and some considerations on how they should be measured. Finally, it describes how fixational eye movements are altered in patients affected by different degrees of vision loss, by movement disorders, or by cognitive disorders.

Keywords Microsaccades · Drift · Nystagmus · Amblyopia · Parkinsonism

1 What Are Fixational Eye Movements and Why Study Them?

Fixational eye movements are the eye movements that occur when subjects attempt to look at a small visual target without letting their gaze move away from it. This task may be referred to as maintained fixation, sustained fixation, visual fixation, or just fixation. During fixation, although with some variability, eye movements are finely controlled to optimize information acquisition (Rolfs 2009; Martinez-Conde et al. 2013; Rucci and Poletti 2015). Fixational eye movements compensate for the spatial and temporal limitations of the retina and the visual system in capturing and

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processing the visual scene. First, the retina has the spatial limitation of not having a homogeneous resolution. Only the fovea has the highest concentration of photoreceptions, and even within the fovea, this concentration varies. Thus, in tasks that require seeing very fine spatial detail, fixational eye movements can align the highest-resolution area of the fovea with the object of interest (Poletti et al. 2013). Second, the visual system has the temporal limitation of neural responses adapting to constant stimuli. Objects may appear to disappear or fade from vision if retinal motion is removed or reduced (Coppola and Purves 1996; Martinez-Conde et al. 2006). This combined action of adaptation of photoreceptors and visual neurons and fixational eye movements can be interpreted in general as a whitening filter that eliminates redundancy in the visual stream and thus optimizes transmission of information (Rucci et al. 2018). Third, the retina also has the temporal limitation of decreased responses to stimuli that change too fast, which presents as perceived smear of rapidly moving objects or reduced visual acuity in the presence of retinal motion (Westheimer and McKee 1975). Thus, eye movements must frequently move the direction of gaze to keep it aligned with the object of interest and to prevent fading and eliminate redundancy in the visual stream while at the same time providing enough periods of relative stability when visual acuity can be optimal. The compromise that the oculomotor system has adopted can be observed not only during fixation but also more generally during visual exploration. The eyes move with an alternation of very quick movements and saccades, interleaved with short periods of stability and fixations.

Visual fixation, even within its simplicity as a task, can provide rich data about multiple brain systems. Performing visual fixation accurately involves vision, cognition, and motor control. Subjects need a healthy visual system, so they can locate and perceive the visual target properly. They need their cognitive abilities to understand the instructions and to continuously maintain fixation on the target. Finally, they need to be able to accurately control their eye muscles to precisely control the direction of gaze. Thus, by measuring fixational eye movements, we can simultaneously obtain information about those three major systems in the brain (Alexander et al. 2018) (Fig. 1).

1.1 Classes of Fixational Eye Movements

Fixational eye movements are usually classified based on descriptive criteria. By looking at the waveforms present in eye movement records, two types of eye movements become obviously distinct: microsaccades as quick steplike changes of eye position and drift as slow smooth eye position changes occurring in between microsaccades. If the recording is made with special devices with high enough resolution (Bolger et al. 1992; McCamy et al. 2013a), a third type of eye movement may be appreciated, tremor, which consists of small fast oscillations that occur simultaneously with drift.

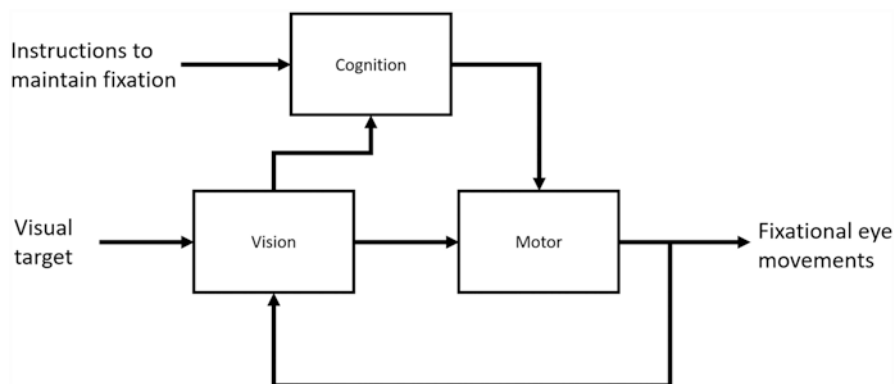


Fig. 1 Schematic representation of the main brain systems involved in fixational eye movements. The visual system is necessary to see the visual stimuli in which to fixate. Cognition is necessary to understand the fixation task and perform it correctly. The oculomotor system should be accurately calibrated to produce the correct eye movements to reach the target and provide a stable position once there

This descriptive classification, however, hides some of the complexity of the oculomotor system. Eye movements can be classified by their specific function, and there is a set of segregated neural structures that correspond with each function. This segregation is of great use in diagnosis and localization of neurological disease, because specific eye movements may be severely deteriorated while others remain intact. The oculomotor system has two main functions, stabilize gaze to reduce retinal image slip and shift gaze to realign the fovea with different parts of the image. These two functions are typically performed in a sequence of alternate states. During the gaze-shifting state, a saccade changes eye position quickly with a minimal role of sensory feedback during the movement. During the image stabilization state, eye movements are controlled by sensory-motor loops in which the appropriate eye movement is continuously monitored using multiple sensory signals to maintain a stable retinal image. The vestibular ocular reflex (VOR) and the optokinetic reflex (OKR) work in concert to generate eye movement that compensates for head movements using vestibular and full-field visual inputs, respectively. Smooth pursuit eye movements are produced to selectively stabilize in the fovea an object moving over a background. The neural integrator generates the necessary drive to the eye muscles to hold stable eye position in the absence of other eye movements. Vergence eye movements occur in combination with saccades, VOR, OKR, and smooth pursuit to generate the appropriate differential eye movements between the two eyes to look at the corresponding position in depth.

While microsaccades can be considered a particular case of saccades that occur during fixation, drift results from the combination of all the other stabilizing movements or even passive movements caused by the elastic properties of the eye. Drift will occur due to noise in each of those systems, miscalibration of their sensory-motor loops or their interrelationships, or abnormal biases or oscillations. Table 1 summarizes all the normal types of eye movements and how they relate to fixational eye movements.

Table 1 Classification of fixational eye movements. NPH: nucleus prepositus hypoglossi, INC: interstitial nucleus of Cajal

Descriptive fixational eye movement type	Description	Functional eye movement type	Main function	Main neural correlate
Microsaccades and/or saccadic intrusions	Quick movements	Saccade	Fast steplike movements to change eye position	Burst of activity in paramedian pontine reticular formation (PPRF) or (horizontal) rostral interstitial nucleus of the medial longitudinal fasciculus (riMLF) (vertical)
		Dynamic overshoot	Fast “bouncing” movement of the eye after a saccade	Burst of activity in PPRF or (horizontal) riMLF (vertical)
Drift	Slow movements	Vestibulo-ocular reflex	Smooth movements to compensate for head movements	Canal-driven activity in the vestibular nuclei
		Optokinetic reflex	Smooth movements to compensate for visual surround movements	Visually driven activity in the vestibular nuclei
		Smooth pursuit	Smooth movements to stabilize a target moving over a surround movement	Activity in the pontine nuclei and flocculus
		Gaze holding	Stable gaze holding at a new position after another eye movement	Sustained activity at NPH (horizontal) and INC (vertical)
		Gaze-evoked drift	An imperfect integrator will tend to drift toward a null position with an exponential decay	Exponential decay of activity in NPH (horizontal) and INC (vertical)
		Glissades	If the gain of the integrator is not well match to its inputs, the eye will drift quickly from the position that ended the previous eye movement to the new	None (passive movement of the eye due to its elastic properties)
		Vergence	Disconjugate movement of the eyes to move and realign the eyes in depth	Vergence neurons in the mesencephalic reticular formation
Tremor	Small fast oscillation			Unknown

2 Methods to Measure and Analyze Fixational Eye Movements

Given the small size of fixational eye movements, it becomes necessary to use eye-tracking devices that have enough temporal and spatial resolution in order to record them and analyze them properly. It is common to use at least 250 Hz and recording systems that can achieve noise levels of 0.1 deg. Currently, the most commonly used family of devices are the infrared video eye trackers which can record up to 2000 Hz. The scleral search coil method (Robinson 1963) is still considered the gold standard providing the best accuracy and precision. Some recent studies have also analyzed fixational eye movements using video recordings of retinal images obtained with an ophthalmoscope (Chung et al. 2015). Some parameters that average data over large windows of time and do not require microsaccade detection may be estimated also in recordings with lower frame rates of 60 Hz, for example, and lower resolution.

2.1 *Microsaccade Measurements*

Analysis of microsaccades starts with a method to identify them and separate them from other movements such as drift or noise that may be present in the recording. Since microsaccades are so small and noise is always present, there is always a compromise between sensitivity and specificity. Typically, a threshold needs to be set, for example, on eye movement velocity to decide if a movement is fast enough to be considered a microsaccade. A too low threshold will result in many false positives and a too high threshold on many false negative or misses. A popular method in the microsaccade field has been developed by Engbert and Kliegl (Engbert and Kliegl 2003) where the threshold is calculated for each recording based on an estimate of the noise level on the data. Still, the method requires a parameter that adjusts the sensitivity (λ), and different studies have used different values ranging from 4 to 8. Recently, a plethora of new methods have been published incorporating novel machine-learning techniques, but none of them have yet become a new standard in the field (Daye and Optican 2014; Otero-Millan et al. 2014; Andersson et al. 2017; Mihali et al. 2017; Bellet et al. 2018; Zemblys et al. 2018). These methods may eliminate the need for the setting of a sensitive parameter and adapt optimally to each patient. They may further provide metrics to evaluate the quality of the recording (Otero-Millan et al. 2014). Further testing is required to compare those methods with different datasets obtained with different recording systems while subjects perform different tasks.

Microsaccade Rate

Microsaccade rate measures the number of microsaccades that occur per unit of time. To measure, one must divide the total number of microsaccades by the total amount of time recorded. When measuring the total amount of time, there are some considerations that may result in slightly different results, for example, whether periods of blinks or missing data are counted toward the total time or not. This can have a large effect on reported rates for subjects with frequent blinks or large amounts of missing data. Many studies have relied in a binocular criterion (only accept microsaccades detected in both eyes) to reduce the number of false positives. Therefore, if two studies use exactly the same detection method with the same parameters but one has only monocular data, it is expected that this study will report higher rates.

Microsaccade Amplitude

Microsaccade amplitude (or magnitude) refers to the size of the movement and the amount of degrees traveled from beginning to end. Two main issues affect microsaccade amplitude measurements. The first is how each study defines amplitude. Some studies will consider the entire excursion of the eye during the movement, while other studies may consider only the distance between the first and the last point of the microsaccade. These two methods will produce different results if dynamic overshoots are common and large. The second is what is the largest amplitude considered. Many studies have defined any saccade smaller than one degree as a microsaccade, while other studies have used more stringent thresholds of half a degree.

Microsaccade Velocity

To measure how fast microsaccades are, one may measure the maximum or peak velocity during the movement. Velocities of saccades and microsaccades are known to follow a parametric relationship with their magnitude called “main sequence” (Bahill et al. 1975). Larger amplitudes tend to be associated with larger velocities, so it is important to dissociate differences in velocities from differences in magnitudes. To fairly compare velocities, one should correct for amplitude first. One possible approach is to fit a regression to the relationship between amplitudes and velocities and report the parameters of such fit. For small saccades including microsaccades, this relationship tends to be linear, so reporting the slope of a linear fit is a good option. For datasets including larger saccades, it is advisable to use some other fit that includes a saturation term and not only a slope, such as the exponential equation $\text{Velocity} = V_{\max} [1 - \exp(-\text{Amplitude}/C)]$, where $V_{\max}$ is the saturation velocity and C determines the shape of the relationship (Rosini et al. 2013).

Microsaccade Direction

Microsaccade direction is usually reported with a graphical representation of their distribution, for example, with a polar plot (Engbert 2006; Otero-Millan et al. 2011). When given summary statistics, one must be careful. Given the circularity of the distribution, it is not appropriate to calculate an overall average direction. For example, in a case where most microsaccades are perfectly horizontal with half to the right and half to the left, an average calculation would result with a vertical direction. Other parameters describing this distribution can still be measured. For example, the ratio between the average absolute value of the horizontal and vertical components will measure if the microsaccade tend to be more horizontal or more vertical.

Microsaccade Intervals

The distribution of intervals of time in between microsaccades can also provide useful information. The average microsaccade rate will correspond mathematically with the inverse of the average time interval between microsaccades. However, there may be more information in the shape of the distribution of the intervals. For example, Otero-Millan and colleagues showed that fitting an ex-Gaussian model (Otero-Millan et al. 2008) to the distributions of saccade intervals, it was possible to observe different parametric relationships with the different parameters of the fit. While different viewing tasks affected the exponential component, the amplitude of the subsequent saccade affected the Gaussian component. Amit and colleagues have recently used a similar analysis to analyze distributions of microsaccade intervals (Amit et al. 2019).

2.2 Drift Parameters

Estimating parameters related to microsaccades is relatively easier than to drift. Microsaccades are short events with high velocities, so during the movement, the signal-to-noise ratio is high. Drift is a much slower and continuous movement which will always have noise added into it and is more likely to be affected by artifacts. For that reason, with many recordings, system may not be possible to assess small differences in drift properties since those may be overwhelmed by the noise. It may still be possible to observe difference in between populations when the effect on drifts is large enough.

The parameters of interest for drift will also be velocity and amplitude, although it is possible also to study its spectral properties to identify oscillations. Because the eye may move back and forth during a given drift period, it is more common to measure variability of both velocity of position.

2.3 *Saccadic Intrusions*

Saccadic intrusions are saccades that intrude or interrupt stable fixation. Recent studies have suggested that saccadic intrusions for a continuum with microsaccades have been given a different name because they have been typically studied in different fields and different populations. Saccadic intrusions in patients are typically larger than in healthy controls and will tend to group together forming specific patterns. Square-wave jerks are the most common type of saccadic intrusions, and they occur when one fixational saccade moves the eye away followed after a short period of stability by another saccade that brings the eye back. Abadi and colleagues have carefully classified the different patterns and measured their frequency and characteristics in healthy controls (Abadi and Gowen 2004).

2.4 *Fixation Stability*

Many studies of the relationship between visual deficits and eye movements have measured fixation stability. Fixation stability measures the overall movement of the eyes while trying to fixate without measuring specific properties of drift and microsaccades. Different studies have measured similar parameters, but one of the most common ones is the bivariate contour ellipse area (BCEA) which measures the dispersion of eye positions in two dimensions. Similar results will be obtained if the standard deviation of eye position or eye velocity is used.

3 **Abnormalities of Fixational Eye Movements**

The next sections review the effects that impaired visual input, motor control, or cognitive function has on fixational eye movements. Here, we will not cover nystagmus, which is characterized by a repetitive eye movement pattern that must include abnormally fast drift. Instead, we will review cases where drift may be abnormally fast but without an obvious repetitive pattern. The presence of nystagmus will pose a challenge when analyzing fixational eye movements. In patients with visual problems, it is common to find latent nystagmus or congenital nystagmus with a wide variety of waveforms. Patients suffering from vestibular disorders or neurological disorders that affect the neural integrator or the vestibular system will typically present with jerk nystagmus (waveform formed by linear or exponential slow phases alternating with quick phases or saccades in the opposite directions). Measurements of drift will be inflated by the nystagmus, and the presence or absence of microsaccades will be confounded with the frequent quick phases of the nystagmus. Thus, when studying populations that may present with nystagmus, it may be advisable to separate those patients into different groups for data analysis (Ghasia et al. 2018).

3.1 Visual Impairments

Eye movements and vision have evolved together. The spatiotemporal characteristics of our visual processing stream depend not only on the properties of the retina but also on the pattern of eye movements that continuously move the image projected onto it (Rucci et al. 2007; Rucci and Poletti 2015). Indeed, there are correlations between the pattern of eye movements that different species make and the anatomy of their visual systems (Samonds et al. 2018). It is not always trivial, however, to establish the causal direction of this correlation. For example, do we continuously move our eyes because the visual system adapts to avoid the fading of images (Martinez-Conde et al. 2006, 2013), or does the visual system adapt because the eyes continuously move to optimally encode visual information? Because of this evolutionary correlation between characteristics of the visual system and characteristics of eye movements, it is expected that problems in the visual system that degrade quality of vision may alter eye movements or, by the same token, that problems in eye movement control may degrade the quality of vision.

In some cases, it is clear that a vision problem is not caused by an eye movement problem such as in macular degeneration. So, we can study how losing vision affects eye movements. In other cases, however, when eye movement abnormalities correlate with vision deficits, it may be more complicated to establish which one is the cause and which one is the effect, such as in amblyopia. This is further complicated by the fact that changes in eye movement output could be directly correlated with the current quality of the visual input, but there could also be other changes that are adaptive and occur over time, thus not relating directly to the current quality of vision but to the history of quality of vision.

Blindness

Visual input is important to maintain accurate and appropriate oculomotor behavior. Blindness, from birth or acquired, causes unstable fixation and possibly nystagmus. Patients with congenital blindness have an impaired vestibulo-ocular reflex and are unable to initiate saccades (Leigh and Zee 1980). Schneider and colleagues (Schneider et al. 2013) studied the effect of monocular and binocular visual loss on the stability of gaze. They showed increased fixation instability with binocular viewing in patients compared to controls, especially in patients with binocular vision loss. They hypothesize that the main effect of vision loss on gaze stability is through the neural integrator based on their results showing more conjugate eye movements during saccades and other eye movements that bypass the neural integrator than during gaze holding. Together, these findings suggest that normal development of some eye movements such as saccades requires vision early on in life. Gaze holding, on the other hand, requires constant calibration by visual input, and acquired visual loss will increase fixation instability.

Amblyopia

Amblyopia refers to a reduction of visual acuity that cannot be explained by refractive errors or by any detectable eye disease (Flom and Neumaier 1966). Amblyopia tends to affect only one eye which is referred to as the amblyopic eye, while the eye with normal vision is called the fellow eye.

When amblyope subjects fixate, looking only with their amblyope eye, they show higher fixation instability measured with BCEA than when viewing with the fellow eye, binocularly, and compared with control (Subramanian et al. 2013; Shaikh et al. 2016). The instability is correlated with the severeness of amblyopia (Shaikh et al. 2016), and it is even larger when amblyopia is related to strabismus (Ghasia et al. 2018).

Increased fixation instability could be caused by increased production of microsaccades, increased drift, or a combination of both or even by microsaccades and drifts that have normal properties but that are coupled in a way that increases the overall instability. Increased drift in amblyopia was first shown by Ciuffreda (Ciuffreda et al. 1980), but recent studies have shown that microsaccades are also altered, having larger amplitude and lower frequency (Shi et al. 2012; Shaikh et al. 2016).

Ghasia and colleagues extended some of the findings related to fixational eye movements to other tasks, such as visual search, and found that the distribution of saccade amplitudes is shifted toward larger amplitudes (Chen et al. 2018). This is consistent with the idea that microsaccades and saccades are part of a continuum (Martinez-Conde et al. 2013; Otero-Millan et al. 2013a) and that the distribution of saccades is adapted to the properties of the visual system (Samonds et al. 2018).

Myopia, Defocus, and Absence of Visual Target

Myopia has also been shown to have an effect on microsaccades. Ghasia and colleagues compared fixational eye movements in subjects with and without correction of their refractive error. They found an increase in microsaccade amplitude with correction of myopia, but microsaccade rate remained unchanged (Ghasia and Shaikh 2015). On the other hand, another recent study found that a simulated refractive error did not cause any change in microsaccade properties (Raveendran et al. 2019). These two findings appear to contradict each other, and it is unclear at the moment why these differences occurred. Moreover, if the fixation spot is completely removed, which could be interpreted as extreme defocus, and subjects are asked to fixate on the center of the screen without a target, microsaccade amplitudes are increased and rates reduced (McCamy et al. 2013b).

3.2 *Movement Control Impairments*

A distributed network across the brain is involved in the accurate control of eye movements. In a simplified model, the frontal cortex, basal ganglia, and superior colliculus (SC) control the initiation of saccades and the inhibition of reflexive saccades, while the cerebellum ensures accuracy of saccades, smooth pursuit, and gaze holding. Impairment in any of these areas will cause a particular pattern of eye movement abnormalities. This entire circuit is also involved in the control of fixation and the generation of microsaccades. Often, abnormalities observed in other eye movements, i.e., slow saccades, will also correspond with the abnormalities present in microsaccades. However, since microsaccades are produced while trying not to move the eyes, there are unique aspects to them that may not be possible to assess by studying other types of eye movements. This might be the case for microsaccade rates and amplitudes.

Among movement disorders, microsaccades have been most studied in parkinsonian syndromes because of the frequent occurrence of saccadic intrusions in these patients that on occasions are visible with the naked eye. In progressive supranuclear palsy (PSP), square-wave jerks are very frequent and large, and they share with saccades their lack of a vertical component and slow velocity (Otero-Millan et al. 2011). In patients with Parkinson's disease (PD), saccade and microsaccade velocities are normal (Bhidayasiri et al. 2001), but the frequency of microsaccades is higher than that in controls (Pinnock et al. 2010; Otero-Millan et al. 2013b). Comparing microsaccade production in PSP and PD, Otero-Millan and colleagues found that while frequency was higher than normal in both groups, the amplitude of microsaccades was larger than normal only in the PSP group (Otero-Millan et al. 2013b).

Recent modeling (Otero-Millan et al. 2018) has shown how the altered properties of microsaccadic amplitude, rate, and velocity in PD and PSP can be simulated by simply modifying two parameters in the model. In this model, microsaccades are generated when fluctuations of activity in the superior colliculus (SC) change the balance of inhibition between omnipause neurons (OPNs) and burst neurons (BNs) in the brain stem. In between microsaccades, OPNs show sustained firing, inhibiting BNs and making them silent. These two populations of neurons inhibit each other, so when the BNs start firing, the circuit behaves as a positive feedback loop with OPNs inhibited by the BNs, thus reducing the inhibition of BNs which further inhibits the OPNs. Lowering the gain of the BNs has the primary effect of reducing the velocities of microsaccades; however, it may also increase the amplitude of microsaccades, since it would take a larger fluctuation of SC activity to break the OPN-BN balance. Increasing the amount of noise that drives the SC fluctuations will have the main effect of increasing the rate of microsaccades. By tweaking these two parameters, it is possible to simulate simultaneously the deficits in three microsaccade properties (velocity, magnitude, and rate) present in PD and PSP populations (Otero-Millan et al. 2013b) (Fig. 2).

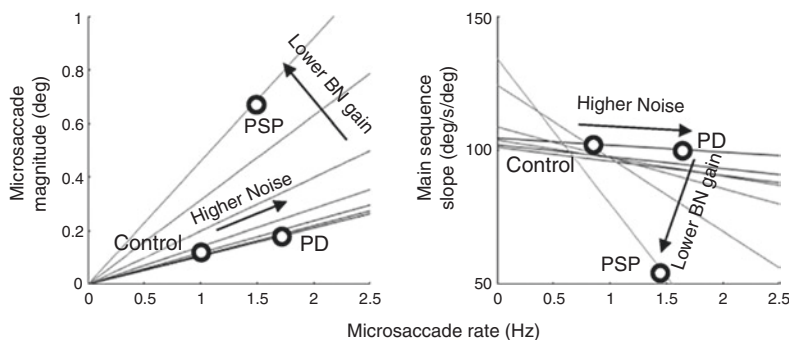


Fig. 2 Modeling of microsaccade production in healthy controls and PSP and PD patients. Simulations of the model varying two parameters (noise level and BN gain) allow to cover the spectrum of microsaccadic rates, velocities, and magnitudes observed on those populations

3.3 Cognitive Impairments

Links between cognition and microsaccade production have been shown in healthy controls. Mental workload induced by asking subjects to perform complicated calculations while fixating on a small target resulted in an increased microsaccade amplitude with reduced frequency (Siegenthaler et al. 2014). Attention also affects microsaccades by biasing their direction sometimes toward and sometimes away from the focus of attention (Hafed and Clark 2002; Engbert and Kliegl 2003; Meyberg et al. 2017). Finally, transient stimulation that can be visual or auditory can temporarily suppress microsaccades (Engbert and Kliegl 2003). It is therefore expected that patients with cognitive impairments could show abnormal microsaccade production.

Fried et al. studied fixational eye movements in attention-deficit hyperactivity disorder (ADHD) patients during a continuous performance test. During the task, subjects can predict when the stimulus is going to appear due to its repetitive timing. Healthy controls tend to suppress their blinks and microsaccades, while ADHD patients don't (Fried et al. 2014). This result, however, may not be generalizable to other tasks like simple visual fixation. Panagiotidi and colleagues studied a non-clinical population of subjects with ADHD traits and found that the score of ADHD traits and microsaccade rate were correlated: subjects with higher scores produced more frequent microsaccades (Panagiotidi et al. 2017).

Alterations in microsaccade in patients with Alzheimer's disease (AD) have also been investigated. While Bylsma and colleagues found an increase in microsaccade rate that correlated with Mini-Mental State Examination (MMSE) score, Kapoula and colleagues did not find a change in microsaccade rate but found that the distribution of microsaccade directions included more oblique microsaccades than healthy controls (Bylsma et al. 1995; Kapoula et al. 2013). In healthy controls, most microsaccades tend to be almost purely horizontal or sometimes vertical, but oblique microsaccades are rare (Otero-Millan et al. 2011).

In autism, the study by Shirama and colleagues found that fixational eye movements were altered in a group of adults with autism spectrum disorder (ASD). They found that instability measured with BCEA was larger in the ASD group than in healthy controls when asked to fixate on the center of the monitor without a fixation target. Under the same conditions, they also found larger and more frequent microsaccades than in controls. When the target was present, their eye movements did not differ between groups (Shirama et al. 2016).

4 Conclusion

The study of fixational eye movements presents opportunities for the development of new diagnostic test and biomarkers that can be used to track progression of patients' efficacy of treatments. There are still some pending questions in the field though. Future studies need to establish the link between vision deficits and fixation stability and what is the contribution of adaptive mechanism over prolonged periods of low vision. It is also entirely unknown what particular oculomotor system is responsible for the increases in fixation stability. Studies on blind patients appear to subject that the major contribution comes from the neural integrator, but this needs to be confirmed in other deficits such as amblyopia.

References

- Abadi, R. V., & Gowen, E. (2004). Characteristics of saccadic intrusions. *Vision Research*, 44, 2675–2690.
- Alexander, R. G., Macknik, S. L., & Martinez-Conde, S. (2018). Microsaccade characteristics in neurological and ophthalmic disease. *Frontiers in Neurology*, 9. Available at: <https://www.frontiersin.org/articles/10.3389/fneur.2018.00144/full>. Accessed 5 Apr 2018.
- Amit, R., Abeles, D., & Yuval-Greenberg, S. (2019). Transient and sustained effects of stimulus properties on the generation of microsaccades. *Journal of Vision*, 19, 6–6.
- Andersson, R., Larsson, L., Holmqvist, K., Stridh, M., & Nyström, M. (2017). One algorithm to rule them all? An evaluation and discussion of ten eye movement event-detection algorithms. *Behav Res*, 49, 616–637.
- Bahill, A. T., Clark, M. R., & Stark, L. (1975). The main sequence, a tool for studying human eye movements. *Mathematical Biosciences*, 24, 191–204.
- Bellet, M. E., Bellet, J., Nienborg, H., Hafed, Z. M., Berens, P. (2018). *Human-level saccade detection performance using deep neural networks*. Available at: <http://biorxiv.org/lookup/doi/10.1101/359018>. Accessed 5 Oct 2018.
- Bhidayasiri, R., Riley, D. E., Somers, J. T., Lerner, A. J., Büttner-Ennever, J. A., & Leigh, R. J. (2001). Pathophysiology of slow vertical saccades in progressive supranuclear palsy. *Neurology*, 57, 2070–2077.
- Bolger, C., Sheahan, N., Coakley, D., & Malone, J. (1992). High frequency eye tremor: Reliability of measurement. *Clinical Physics and Physiological Measurement*, 13, 151–159.

- Bylsma, F. W., Rasmussen, D. X., Rebok, G. W., Keyl, P. M., Tune, L., & Brandt, J. (1995). Changes in visual fixation and saccadic eye movements in Alzheimer's disease. *International Journal of Psychophysiology*, 19, 33–40.
- Chen, D., Otero-Millan, J., Kumar, P., Shaikh, A. G., & Ghasia, F. F. (2018). Visual search in amblyopia: Abnormal fixational eye movements and suboptimal sampling strategies. *Investigative Ophthalmology & Visual Science*, 59, 4506–4517.
- Chung, S. T. L., Kumar, G., Li, R. W., & Levi, D. M. (2015). Characteristics of fixational eye movements in amblyopia: Limitations on fixation stability and acuity? *Vision Research*, 114, 87–99.
- Ciuffreda, K. J., Kenyon, R. V., & Stark, L. (1980). Increased drift in amblyopic eyes. *British Journal of Ophthalmology*, 64, 7–14.
- Coppola, D., & Purves, D. (1996). The extraordinarily rapid disappearance of entopic images. *Proceedings of the National Academy of Sciences*, 93, 8001–8004.
- Daye, P. M., & Optican, L. M. (2014). Saccade detection using a particle filter. *Journal of Neuroscience Methods*, 235, 157–168.
- Engbert, R. (2006). Microsaccades: a microcosm for research on oculomotor control, attention, and visual perception. In *Progress in brain research* (pp. 177–192). Elsevier. Available at: <http://linkinghub.elsevier.com/retrieve/pii/S0079612306540099>. Accessed 6 July 2015.
- Engbert, R., & Kliegl, R. (2003). Microsaccades uncover the orientation of covert attention. *Vision Research*, 43, 1035–1045.
- Flom, M. C., & Neumaier, R. W. (1966). Prevalence of amblyopia. *Public Health Reports*, 81, 329–341.
- Fried, M., Tsitsiashvili, E., Bonne, Y. S., Sterkin, A., Wygnanski-Jaffe, T., Epstein, T., & Polat, U. (2014). ADHD subjects fail to suppress eye blinks and microsaccades while anticipating visual stimuli but recover with medication. *Vision Research*, 101, 62–72.
- Ghasia, F. F., & Shaikh, A. G. (2015). Uncorrected myopic refractive error increases microsaccade amplitude. *Investigative Ophthalmology & Visual Science*. Available at: <http://www.iovs.org/cgi/doi/10.1167/iovs.14-15882>. Accessed 2 June 2015, 56, 2531.
- Ghasia, F. F., Otero-Millan, J., & Shaikh, A. G. (2018). Abnormal fixational eye movements in strabismus. *British Journal of Ophthalmology*, 102, 253–259.
- Hafed, Z. M., & Clark, J. J. (2002). Microsaccades as an overt measure of covert attention shifts. *Vision Research*, 42, 2533–2545.
- Kapoula, Z., Yang, Q., Otero-Millan, J., Xiao, S., Macknik, S. L., Lang, A., Verny, M., & Martinez-Conde, S. (2013). Distinctive features of microsaccades in Alzheimer's disease and in mild cognitive impairment. *Age (Dordr)*, 36(2), 535–543.
- Leigh, R. J., & Zee, D. S. (1980). Eye movements of the blind. *Investigative Ophthalmology & Visual Science*, 19, 328–331.
- Martinez-Conde, S., Macknik, S. L., Troncoso, X. G., & Dyar, T. A. (2006). Microsaccades counteract visual fading during fixation. *Neuron*, 49, 297–305.
- Martinez-Conde, S., Otero-Millan, J., & Macknik, S. L. (2013). The impact of microsaccades on vision: Towards a unified theory of saccadic function. *Nature Reviews Neuroscience*, 14, 83–96.
- McCamy, M. B., Collins, N., Otero-Millan, J., Al-Kalbani, M., Macknik, S. L., Coakley, D., Troncoso, X. G., Boyle, G., Narayanan, V., Wolf, T. R., & Martinez-Conde, S. (2013a). Simultaneous recordings of ocular microtremor and microsaccades with a piezoelectric sensor and a video-oculography system. *PeerJ*, 1, e14.
- McCamy, M. B., Najafian Jazi, A., Otero-Millan, J., Macknik, S. L., & Martinez-Conde, S. (2013b). The effects of fixation target size and luminance on microsaccades and square-wave jerks. *PeerJ*, 1, e9.
- Meyberg, S., Sinn, P., Engbert, R., & Sommer, W. (2017). Revising the link between microsaccades and the spatial cueing of voluntary attention. *Vision Research*, 133, 47–60.
- Mihali, A., Opheusden, B. V., & Ma, W. J. (2017). Bayesian microsaccade detection. *Journal of Vision*, 17, 13–13.

- Otero-Millan, J., Troncoso, X. G., Macknik, S. L., Serrano-Pedraza, I., & Martinez-Conde, S. (2008). Saccades and microsaccades during visual fixation, exploration and search: Foundations for a common saccadic generator. *Journal of Vision*, 8(14), 21.
- Otero-Millan, J., Serra, A., Leigh, R. J., Troncoso, X. G., Macknik, S. L., & Martinez-Conde, S. (2011). Distinctive features of saccadic intrusions and microsaccades in progressive supranuclear palsy. *Journal of Neuroscience*, 31, 4379–4387.
- Otero-Millan, J., Macknik, S. L., Langston, R. E., & Martinez-Conde, S. (2013a). An oculomotor continuum from exploration to fixation. *PNAS*, 110, 6175–6180.
- Otero-Millan, J., Schneider, R., Leigh, R. J., Macknik, S. L., & Martinez-Conde, S. (2013b). Saccades during attempted fixation in parkinsonian disorders and recessive Ataxia: From microsaccades to square-wave jerks Geng JJ, ed. *PLoS One*, 8, e58535.
- Otero-Millan, J., Castro, J. L. A., Macknik, S. L., & Martinez-Conde, S. (2014). Unsupervised clustering method to detect microsaccades. *Journal of Vision*, 14, 18–18.
- Otero-Millan, J., Optican, L. M., Macknik, S. L., & Martinez-Conde, S. (2018). Modeling the triggering of saccades, microsaccades, and saccadic intrusions. *Frontiers in Neurology*, 9. Available at: <https://www.frontiersin.org/articles/10.3389/fneur.2018.00346/full>. Accessed 18 July 2018.
- Panagiotidi, M., Overton, P. G., & Stafford, T. (2017). Increased microsaccade rate in individuals with ADHD traits. *Journal of Eye Movement Research*, 10. Available at: <https://doi.org/10.16910/jemr.10.1.6>. Accessed 4 Oct 2017.
- Pinnock, R. A., McGivern, R. C., Forbes, R., & Gibson, J. M. (2010). An exploration of ocular fixation in Parkinson's disease, multiple system atrophy and progressive supranuclear palsy. *Journal of Neurology*, 257, 533–539.
- Poletti, M., Listorti, C., & Rucci, M. (2013). Microscopic eye movements compensate for nonhomogeneous vision within the fovea. *Current Biology*, 23, 1691–1695.
- Raveendran, R. N., Bobier, W., & Thompson, B. (2019). Reduced amblyopic eye fixation stability cannot be simulated using retinal-defocus-induced reductions in visual acuity. *Vision Research*, 154, 14–20.
- Robinson, D. A. (1963). A method of measuring eye movements using a scleral coil in a magnetic field. *IEEE Transactions on Biomedical Engineering*, 10, 137–145.
- Rolfs, M. (2009). Microsaccades: Small steps on a long way. *Vision Research*, 49, 2415–2441.
- Rosini, F., Federighi, P., Pretelegiani, E., Piu, P., Leigh, R. J., Serra, A., Federico, A., & Rufa, A. (2013). Ocular-motor profile and effects of memantine in a familial form of adult cerebellar ataxia with slow saccades and square wave saccadic intrusions Martinez-Conde S, ed. *PLoS One*, 8, e69522.
- Rucci, M., & Poletti, M. (2015). Control and functions of fixational eye movements. *Annual Review of Vision Science*, 1, 499–518.
- Rucci, M., Iovin, R., Poletti, M., & Santini, F. (2007). Miniature eye movements enhance fine spatial detail. *Nature*, 447, 852–855.
- Rucci, M., Ahissar, E., & Burr, D. (2018). Temporal coding of visual space. *Trends in Cognitive Sciences*, 22, 883–895.
- Samonds, J. M., Geisler, W. S., & Priebe, N. J. (2018). Natural image and receptive field statistics predict saccade sizes. *Nature Neuroscience*, 21, 1591–1599.
- Schneider, R. M., Thurtell, M. J., Eisele, S., Lincoff, N., Bala, E., & Leigh, R. J. (2013). Neurological basis for eye movements of the blind. *PLoS One*, 8, e56556.
- Shaikh, A. G., Otero-Millan, J., Kumar, P., & Ghasia, F. F. (2016). Abnormal fixational eye movements in amblyopia. *PLoS One*, 11, e0149953.
- Shi, X. F., Xu, L., Li, Y., Wang, T., Zhao, K., & Sabel, B. A. (2012). Fixational saccadic eye movements are altered in anisometric amblyopia. *Restorative Neurology and Neuroscience*, 30, 445–462.
- Shirama, A., Kanai, C., Kato, N., & Kashino, M. (2016). Ocular fixation abnormality in patients with autism spectrum disorder. *Journal of Autism and Developmental Disorders*, 46, 1613–1622.

- Siegenthaler, E., Costela, F. M., McCamy, M. B., Di Stasi, L. L., Otero-Millan, J., Sonderegger, A., Groner, R., Macknik, S., & Martinez-Conde, S. (2014). Task difficulty in mental arithmetic affects microsaccadic rates and magnitudes. *European Journal of Neuroscience*, 39, 287–294.
- Subramanian, V., Jost, R. M., & Birch, E. E. (2013). A quantitative study of fixation stability in amblyopia. *Investigative Ophthalmology & Visual Science*, 54, 1998–2003.
- Westheimer, G., & McKee, S. P. (1975). Visual acuity in the presence of retinal-image motion. *JOSA*, 65, 847–850.
- Zemblys, R., Niehorster, D. C., Komogortsev, O., & Holmqvist, K. (2018). Using machine learning to detect events in eye-tracking data. *Behavior Research Methods*, 50, 160–181.